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Course Code: DSA0503

Course Name: Query Processing for Data Science

Assessment: C01-AT01

Question:

Parse the JSON structured text block containing global city metrics into a Pandas DataFrame. Display only the specific columns: city, temperature, humidity, and wind_speed.

Aim:

To write a Python program using the Pandas library to parse a JSON weather dataset, display specific environmental metrics (City Name, Temperature, Humidity, and Wind Speed), and programmatically identify the city recording the maximum temperature.

Algorithm

1. Start the program.
2. Import the json module.
3. Store the weather information in JSON format.
4. Convert the JSON string into Python objects using json.loads().
5. Initialize the first city as the hottest city.
6. Traverse each weather record in the JSON data.
7. Display the following details for every city:
 - City Name
 - Temperature
 - Humidity
 - Wind Speed
8. Compare the temperature of the current city with the hottest city's temperature.
9. If the current city's temperature is higher, update the hottest city.
10. After checking all cities, display the hottest city's name and temperature.

JSON:

```
[
{
  "city": "Hyderabad",
  "temperature": 38,
  "humidity": 60,
```

```
"wind_speed": 10
},

{
  "city": "Bengaluru",
  "temperature": 30,
  "humidity": 75,
  "wind_speed": 15
},

{
  "city": "Delhi",
  "temperature": 42,
  "humidity": 45,
  "wind_speed": 8
},

{
  "city": "Kolkata",
  "temperature": 36,
  "humidity": 78,
  "wind_speed": 14
},

{
  "city": "Pune",
  "temperature": 31,
  "humidity": 65,
  "wind_speed": 11
},

{
  "city": "Ahmedabad",
  "temperature": 40,
  "humidity": 50,
  "wind_speed": 9
},

{
  "city": "Jaipur",
  "temperature": 39,
  "humidity": 40,
  "wind_speed": 13
```

```

    },

    {
        "city": "Visakhapatnam",
        "temperature": 34,
        "humidity": 76,
        "wind_speed": 16
    }
]

```

Python Code:

```

import json

data = '''

[
    {"city":"Hyderabad","temperature":38,"humidity":60,"wind_speed":10},
    {"city":"Bengaluru","temperature":30,"humidity":75,"wind_speed":15},
    {"city":"Delhi","temperature":42,"humidity":45,"wind_speed":8},
    {"city":"Kolkata","temperature":36,"humidity":78,"wind_speed":14},
    {"city":"Pune","temperature":31,"humidity":65,"wind_speed":11},
    {"city":"Ahmedabad","temperature":40,"humidity":50,"wind_speed":9},
    {"city":"Jaipur","temperature":39,"humidity":40,"wind_speed":13},
    {"city":"Visakhapatnam","temperature":34,"humidity":76,"wind_speed":16}
] '''

weather = json.loads(data)

hottest = weather[0]

print("Weather Report")

print("-" * 50)

for city in weather:

    print(f'City      : {city["city"]}')

    print(f'Temperature : {city["temperature"]} °C')

    print(f'Humidity    : {city["humidity"]} %')

    print(f'Wind Speed   : {city["wind_speed"]} km/h')

```

```

print("-" * 50)

if city["temperature"] > hottest["temperature"]:

    hottest = city

print("\nHottest City")

print(f'City      : {hottest['city']}')

print(f'Temperature : {hottest['temperature']} °C")

```

Output:

```

IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: D:\Query Processing\Assignment -1.py =====
Weather Report
-----
City      : Hyderabad
Temperature : 38 °C
Humidity   : 60 %
Wind Speed : 10 km/h
-----
City      : Bengaluru
Temperature : 30 °C
Humidity   : 75 %
Wind Speed : 15 km/h
-----
City      : Delhi
Temperature : 42 °C
Humidity   : 45 %
Wind Speed : 8 km/h
-----
City      : Kolkata
Temperature : 36 °C
Humidity   : 78 %
Wind Speed : 14 km/h
-----
City      : Pune
Temperature : 31 °C
Humidity   : 65 %
Wind Speed : 11 km/h
-----
City      : Ahmedabad
Temperature : 40 °C
Humidity   : 50 %
Wind Speed : 9 km/h
-----
City      : Jaipur
Temperature : 39 °C
Humidity   : 40 %
Wind Speed : 13 km/h
-----
City      : Visakhapatnam
Temperature : 34 °C
Humidity   : 76 %
Wind Speed : 16 km/h
-----
Hottest City
City      : Delhi
Temperature : 42 °C
>>>

```

The program successfully displayed the weather details of all cities and identified Delhi as the hottest city with a temperature of 42°C

Result:

The JSON weather data was successfully read and displayed. The city name, temperature, humidity, and wind speed for each city were printed correctly. By comparing the temperatures of all cities, the program successfully identified the hottest city.